

Does Idiomatic Phrasing Degrade LLM Task Performance? A Controlled Paraphrase-vs-Idiom Study Across Instruction-Tuned Decoder Models

Assaf Haik Barouch (206783441) Lihu Zur (322216151)

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1 Introduction

Idiomatic expressions — those whose meaning cannot be derived compositionally from their individual words, e.g., “*spill the beans*” or “*par for the course*” — are routinely treated in NLP as a harder case for language models than direct, literal phrasing. The premise is intuitive: idioms are non-compositional, culturally specific, and comparatively rare in balanced training data relative to literal phrasing of the same content. It is, however, rarely tested *in isolation*: any two rewordings of a sentence already differ lexically and syntactically, so an observed accuracy drop after “idiomatizing” it could simply be an artifact of rewording in general.

Central research question: does phrasing a task input idiomatically cause a measurable degradation in a language model’s task accuracy, *over and above* the degradation that any comparable paraphrase (a non-idiomatic rewording) already causes?

To answer this cleanly, we rewrite every original example x into two controlled variants that share the same rewriting budget but differ in exactly one property: $x_{\text{paraphrase}}$, reworded in different words **without** idioms, which is the *control* condition isolating the effect of rewording alone; and $x_{\text{idiomatic}}$, reworded **with** one or more natural idiomatic expressions and the same meaning. The label y is preserved across all three variants by construction (§2.2), and each variant is evaluated under an identical prompt template, so only the surface form of the input changes between runs. This paired design lets us attribute degradation seen under the idiomatic variant beyond what the paraphrase variant already causes to idiomaticity specifically.

The question matters practically, because idiomatic language is common in the user-generated text deployed systems must handle. If idiomaticity has an isolable negative effect, that bears on prompt engineering, augmentation strategy, and robustness evaluation; if apparent idiom failures are merely generic paraphrase sensitivity, then effort spent specifically on idiom handling is misplaced relative to general robustness-to-rewording work.

Contributions. We contribute (i) a paraphrase-controlled three-way design, in which the paraphrase arm absorbs the generic cost of rewording and leaves the idiom-specific effect isolated; (ii) a *sole-cause* diagnostic that, restricted to items a model already answers correctly, identifies which single rewrite is responsible when exactly one of the two flips the answer; (iii) dual significance testing, pairing a continuity-corrected McNemar test with an independent non-parametric bootstrap to guard against the chi-square approximation on small discordant-pair counts; (iv) breadth — seventeen instruction-tuned decoder LLMs from 0.5B to 9B parameters across eleven model families, on three datasets spanning sentiment, knowledge QA, and inference, for 51 (model, dataset) runs with a pooled cross-model test per dataset; and (v) a reproducible four-stage pipeline (clean → augment → evaluate → analyze) with LLM-judge validation of every augmented example and per-artifact provenance (§2).

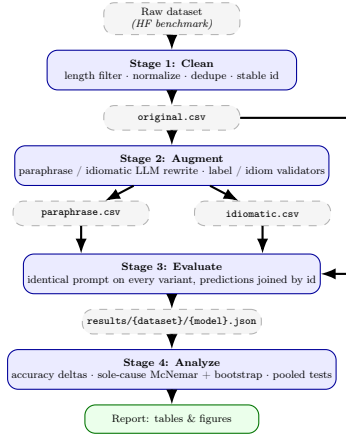


Figure 1: The four-stage pipeline (the stage orchestrators carry no dataset- or model-specific branching; each dataset adds its own loader and evaluator module). The same pipeline instantiates once per dataset/model combination. Raw data is cleaned (Stage 1), rewritten into validated paraphrase and idiomatic variants (Stage 2), evaluated per model across every variant with an identical prompt (Stage 3), and aggregated into the accuracy-delta and sole-cause significance analysis (Stage 4). `original.csv` feeds both Stage 2 (as the rewriting source) and Stage 3 (as the control variant) directly.

1.1 Related Work

Idiom processing in NLP. Prior work has mostly focused on *detecting* or *translating* idioms rather than measuring their effect on downstream accuracy. Sporleder and Li [1]¹ classify literal versus idiomatic usage of an expression in context; Tayyar Madabushi et al. [2]² probe pretrained models on the same distinction, finding they do reasonably well given one or a few labelled examples per expression but degrade substantially zero-shot on unseen expressions; and Chakrabarty et al. [3]³ introduce FLUTE, a 9,000-instance figurative-language benchmark spanning idioms, similes, metaphors, and sarcasm. These establish that idiom *comprehension* is a genuine weak point, but do not isolate how much of an idiom’s effect on a *downstream task* is attributable to idiomaticity rather than to rewording in general.

Paraphrase and rewording robustness. A separate line of work shows models are sensitive to input edits that leave the correct answer intact: Jia and Liang [4]⁴ fool reading-comprehension models by *appending* distractor sentences that do not change the answer, and Ribeiro et al. [5]⁵ flip predictions across several tasks by *rewording* the input under semantics-preserving rules (SEARs). This is why a bare original-versus-idiomatic comparison is confounded: any degradation could be this generic phenomenon rather than anything specific to idioms. To our knowledge no prior work isolates idiom-specific from generic paraphrase-induced degradation via a paired three-way control design across a broad panel of contemporary open instruction-tuned decoder LLMs, which is the gap this work addresses.

2 Methodology

2.1 Pipeline overview

The project is implemented as four independently re-runnable stages (Figure 1) that communicate only through versioned files on disk, plus YAML configs that set each stage’s run parameters; dataset-specific loading and answer-parsing logic lives in per-dataset modules. Any stage can therefore be re-executed alone — evaluating a new model does not require repeating cleaning or augmentation — and every artifact records the config hash and model/prompt version that produced it.

¹[1] Sporleder and Li (2009), Literal and Non-Literal Use of Idiomatic Expressions, EACL.

²[2] Tayyar Madabushi et al. (2021), ASitchInLanguageModels: Idiomaticity in Pre-Trained LMs, Findings of EMNLP.

³[3] Chakrabarty et al. (2022), FLUTE: Figurative Language Understanding through Textual Explanations, EMNLP.

⁴[4] Jia and Liang (2017), Adversarial Examples for Evaluating Reading Comprehension, EMNLP.

⁵[5] Ribeiro et al. (2018), Semantically Equivalent Adversarial Rules for Debugging NLP Models, ACL.

Stage 1: Clean. Each raw dataset is loaded from Hugging Face and reduced to one clean CSV of rows suitable for rewriting: rows outside a configured length band are dropped (very short inputs leave no room for an idiom; very long ones are expensive to augment and dilute the measured effect), text is normalized (Unicode NFC, whitespace collapsing), exact duplicates are removed, and every retained row gets a stable identifier that aligns all its downstream variants. The label y and dataset-specific metadata (e.g., MMLU’s answer choices) carry through unchanged.

2.2 Augmentation into paraphrase and idiomatic variants

Stage 2: Augment. Each cleaned row is rewritten by an external LLM (Google Gemini) into two variants using two distinct, fixed prompt templates. Templates are selected per dataset, so a task whose input is not a free-standing sentence can use wording suited to it (MNLI, whose rewrite target is a premise judged against a fixed hypothesis, uses its own NLI-specific pair of templates) without perturbing the prompt hashes, and therefore the warm caches, of the other datasets:

the *paraphrase prompt* asks for a rewording in different words that explicitly avoids idiomatic expressions, and the *idiomatic prompt* asks for one that naturally incorporates one or more idioms or figures of speech while preserving meaning, register, and length. Both are given the task label y in the prompt itself, so a rewrite cannot silently invert the correct answer, e.g., flipping a positive sentiment snippet into a negative one. An external, deliberately stronger LLM is used rather than one of the models under evaluation, since the augments’s own idiom/paraphrase judgment must be more reliable than the systems being tested.

Every augmented row then passes a validator chain: **label preservation**, an LLM-judge check that the rewrite has not altered the correct label; and **idiom presence / absence**, a per-variant LLM-judge check requiring a genuine idiom in the idiomatic variant and none in the paraphrase. Rows that fail validation after a bounded number of retries are dropped from that variant (recorded in a sidecar manifest for transparency), which is why the paraphrase/idiomatic CSVs are typically somewhat smaller than the original CSV. An on-disk response cache, keyed by (prompt hash, augments model, input id), makes reruns idempotent and avoids re-incurring API cost for rows that have already been successfully augmented under an unchanged configuration.

2.3 Datasets

Three datasets are evaluated, each scored with the accuracy metric used in its original publication. Stage 2 rewrites one field per dataset — SST-2’s sentence, MMLU’s question stem, MNLI’s premise — and everything else, including MMLU’s answer choices and MNLI’s hypothesis, stays byte-identical across the three variants.

SST-2 [6]⁶ is binary sentiment classification over short movie-review snippets, loaded from the GLUE release [7]⁷; **MMLU** [8]⁸ is multiple-choice knowledge questions spanning 57 subjects; and **MNLI** [9]⁹ is three-way inference (*entailment* / *neutral* / *contradiction*) over premise–hypothesis pairs from ten written and spoken genres.

The evaluated split differs by dataset according to what each release makes available with labels: MMLU uses its **validation** split and MNLI its two validation splits (below), while SST-2, whose GLUE test labels are withheld, draws from **train** and **validation** combined, capped at 2,000 rows.

The three cover qualitatively different task types under one protocol: short, informal, already-somewhat-figurative text; longer, formal, fact-oriented multiple-choice questions; and sentence-pair inference. MNLI is the most direct test of the three, because in sentiment or knowledge QA an idiom perturbs the surface of an input whose answer is otherwise independent of it, whereas an entailment judgement turns on a literal reading of the premise itself. Its three-way label space also sets chance at 33% rather

⁶[6] Socher et al. (2013), Recursive Deep Models over a Sentiment Treebank, EMNLP.

⁷[7] Wang et al. (2019), GLUE: A Multi-Task Benchmark for NLU, ICLR.

⁸[8] Hendrycks et al. (2021), Measuring Massive Multitask Language Understanding, ICLR.

⁹[9] Williams et al. (2018), A Broad-Coverage Challenge Corpus for Sentence Understanding through Inference, NAACL.

than 50%, which depresses the per-row rate of sole-cause events relative to a binary task.

MNLI-specific protocol. MNLI pairs roughly three hypotheses with every premise; emitting all of them would repeat the same premise and violate the independence assumption behind the paired test, so the loader keeps only the first hypothesis per premise. Rows come from the two validation splits combined (matched, 9,815 pairs; mismatched, 9,832), the test labels being withheld. Because the mismatched half contributes the five genres absent from training, each row’s genre is recorded so that genre shift stays visible as a confound. After first-hypothesis selection and cleaning, 5,127 premises survive to form the MNLI original file, of which 4,709 (91.8%) additionally clear validation in *both* rewritten variants and so form the aligned triples the analysis runs on.

2.4 Evaluated models

Rather than a single-model case study, we evaluate a broad panel of seventeen open, instruction-tuned decoder LLMs spanning 0.5B to 9B parameters across eleven distinct model families and pretraining lineages: the Qwen2.5-Instruct family (0.5B / 1.5B / 3B / 7B), the Gemma-2-it family (2B / 9B), the Falcon3-Instruct family (3B / 7B), the OLMo-2-Instruct family (1B / 7B), Mistral-7B-Instruct-v0.3, Phi-4-mini-instruct (3.8B), Granite-3.1-2B-Instruct, H2O-Danube3-4B-Chat, Yi-1.5-6B-Chat, SmolLM2-1.7B-Instruct, and StableLM-2-1.6B-Chat. Spanning multiple scales *within* a family (four Qwen2.5 sizes) as well as different families at similar scale lets the analysis separate size-driven from family-driven effects rather than conflating them, as a single family swept across sizes would. All models are evaluated zero-shot with an identical dataset-specific prompt template applied to all three variants of a row, so only the input text differs between runs; a device-agnostic runner selects CUDA, Apple-Silicon MPS, or CPU and adjusts precision accordingly.

Stage 3: Evaluate. For each (model, dataset) pair, the model is run on all three variant CSVs with the identical prompt, and predictions are joined by row id, yielding one matched (original, paraphrase, idiomatic) triple of predictions per surviving row. Per-variant accuracy is computed directly from these joined predictions.

2.5 Statistical framework

Stage 4: Analyze. Two families of statistic are computed. The first is per-variant accuracy and the two deltas between variants, reported throughout in percentage points (pp):

$$\Delta_{\text{par}} = \text{acc}_{\text{par}} - \text{acc}_{\text{orig}}, \quad \Delta_{\text{idi}} = \text{acc}_{\text{idi}} - \text{acc}_{\text{par}}. \quad (1)$$

Δ_{par} measures the cost of rewording and Δ_{idi} the additional cost of idiomaticity on top of it; the research question of §1 is precisely whether $\Delta_{\text{idi}} < 0$. Each is accompanied by a paired-bootstrap confidence interval and a McNemar exact p-value over the corresponding discordant pairs.

The second, and our primary diagnostic, is a *sole-cause* test, which attributes degradation to a single rewrite type rather than to ordinary noise. It restricts attention to rows where all three variants were produced and the model answered the **original** correctly — so any subsequent error is attributable to the rewrite, not to the item being hard for that model — and then counts $b = \textit{idiomatic-only-wrong}$ (the idiomatic rewrite alone broke an otherwise-correct answer) and $c = \textit{paraphrase-only-wrong}$ (the paraphrase rewrite alone broke it). Rows where both rewrites agree, breaking it or not, carry no information about which is worse and are excluded, exactly as in any paired test. The gap between b and c is then assessed two ways:

McNemar’s test [10]¹⁰, the standard test for paired nominal before/after comparisons, with the continuity correction of Edwards [11]¹¹:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}, \quad \text{df} = 1, \quad (2)$$

¹⁰[10] [McNemar \(1947\)](#), Sampling Error of the Difference Between Correlated Proportions, Psychometrika.

¹¹[11] [Edwards \(1948\)](#), Correction for Continuity in Correlated Proportions, Psychometrika.

Table 1: Per-dataset summary over the aligned variant triples. Δ values are means across models in percentage points; *sig.* counts models with a significant negative / positive Δ_{idi} ($p < 0.05$, McNemar exact). *Parse-clean* repeats both after dropping every triple in which any variant failed to parse, since an unparseable generation is scored as incorrect. The pooled sole-cause columns sum b (idiomatic-only-wrong) and c (paraphrase-only-wrong) across models and test the gap with continuity-corrected McNemar and a 10,000-resample paired bootstrap.

Dataset	Models	n	All triples			Parse-clean		Pooled sole-cause			
			Δ_{par}	Δ_{idi}	sig.	Δ_{idi}	sig.	b / c	χ^2	p_{χ^2}	p_{boot}
SST-2	17	1,548	-1.25	+1.31	0 / 5	+0.67	0 / 3	665 / 887	31.5	<0.0001	<0.0001
MMLU	17	1,085	-0.07	-0.71	1 / 0	-0.73	1 / 0	578 / 468	11.4	0.0008	0.0012
MNLI	17	4,709	-0.69	-1.16	10 / 0	-1.14	10 / 0	3351 / 2751	58.8	<0.0001	<0.0001

Table 2: Per-model Δ_{idi} (pp) on each dataset, ordered by parameter count and split into two panels. **Bold** marks Δ_{idi} significant at $p < 0.05$ (McNemar exact); $\dagger$ marks models whose sole-cause gap between b and c is significant on *both* the McNemar and bootstrap tests, in whichever direction. Negative values support the hypothesis that idiomatic phrasing costs accuracy beyond paraphrasing; note that the SST-2 columns are almost entirely positive.

Model	SST-2	MMLU	MNLI	Model	SST-2	MMLU	MNLI
Qwen2.5-0.5B	+0.45	-1.75	-0.40	Phi-4-mini-3.8B	-0.19	-1.84	- 1.51 $\dagger$
OLMo-2-1B	+1.29	-0.28	- 1.70 $\dagger$	Danube3-4B	+ 1.74	- 2.49	+0.04
Qwen2.5-1.5B	+1.16	-2.03 $\dagger$	- 1.59 $\dagger$	Yi-1.5-6B	+0.97	-0.18	- 1.89 $\dagger$
StableLM-2-1.6B	+1.16	-0.28	-0.15	Qwen2.5-7B	+ 3.04 $\dagger$	+0.09	- 1.17 $\dagger$
SmolLM2-1.7B	+ 2.20	-0.74	+0.30	Falcon3-7B	-0.26	+0.65	- 1.23 $\dagger$
Gemma-2-2B	+1.36 $\dagger$	-1.38	- 1.51 $\dagger$	Mistral-7B-v0.3	+ 4.13 $\dagger$	-1.20	- 2.08
Granite-3.1-2B	+ 1.49 $\dagger$	+2.12	-0.89	OLMo-2-7B	+1.16	-1.66 $\dagger$	- 2.04 $\dagger$
Qwen2.5-3B	+0.84	+0.92	-0.85	Gemma-2-9B	+1.29	-1.66	-0.85
Falcon3-3B	+0.39	-0.37	- 2.25 $\dagger$				

testing the null hypothesis that b and c are drawn from the same underlying 50/50 rate (i.e., among discordant rows, it is a coin flip which rewrite broke it).

Non-parametric bootstrap check. Because that chi-square approximation can be unreliable when $b + c$ is small, every result is cross-checked with a paired bootstrap [12, 13]^{12,13}: the aligned rows are resampled with replacement 10,000 times, $b - c$ is recomputed on each resample, and a 95% percentile interval plus a two-sided empirical p-value are read off the result. A result counts as robustly significant only when both checks agree; when they disagree the more conservative verdict is preferred.

The same sole-cause statistics are additionally pooled across all evaluated models for a given dataset (summing b and c , and resampling from the concatenated pooled rows for the bootstrap), with the explicit caveat that this pooling is not a rigorous mixed-effects test, since different models are not strictly exchangeable independent draws, and is reported as indicative rather than definitive.

3 Experimental results

The sweep covers 51 (model, dataset) runs — all seventeen models on each of the three datasets — and so 153 variant-level inference passes, about 34 hours of accelerator time. Every number below is computed over *aligned triples*: rows for which the original and both validated rewrites exist and the model produced a scored prediction on all three, leaving 1,548 rows for SST-2, 1,085 for MMLU and 4,709 for MNLI. Stage 3 also evaluates the full original files (2,000 / 1,107 / 5,127 rows), but those rows have no counterpart to pair against.

The effect is task-dependent, not general (Table 1). On MNLI the hypothesis holds — Δ_{idi} averages -1.16 pp, significantly negative for 10 of 17 models and positive for none, with pooled sole-cause counts of $b = 3,351$ against $c = 2,751$ ($p < 10^{-4}$ on both tests). On MMLU it holds weakly and only in aggregate, where just one model reaches individual significance but the pooled test does ($p = 0.0008$

¹²[12] Efron and Tibshirani (1993), An Introduction to the Bootstrap, Chapman & Hall.

¹³[13] Koehn (2004), Statistical Significance Tests for MT Evaluation, EMNLP.

McNemar, $p = 0.001$ bootstrap). On SST-2 it is contradicted: the mean is +1.31 pp, five models are significantly *positive* and none negative, and the pooled sole-cause test is significant in the opposite direction. On MNLI, moreover, idiomaticity costs more than the rewording carrying it (−1.16 against −0.69 pp), whereas on SST-2 the two nearly cancel.

SST-2: rewording hurts, idioms recover. Original accuracy is high and tightly clustered (83.8–91.8%), and plain paraphrasing costs accuracy for 14 of 17 models, significantly for 6. Idiomatizing that same rewording recovers most of the loss — positive for 15 of 17 models, significantly so for 5 — with Mistral-7B-v0.3 showing the largest swing, −2.91 pp from paraphrasing and +4.13 pp from idiomatizing. The parse-failure control below shows, however, that part of this result is artifact rather than comprehension.

MMLU: a small effect that survives only pooling. MMLU spreads models much more widely (29.7–72.3%, against a 25% chance floor) and both deltas are small: Δ_{par} averages −0.07 pp with no model significantly degraded, Δ_{idi} −0.71 pp with only Danube3-4B individually significant. The sole-cause direction is consistent enough ($b > c$ for 12 of 17 models) that pooling turns it into a clear signal. The small magnitude is likely structural: only the stem is rewritten, while the answer choices — where an MMLU item’s knowledge requirement sits — are untouched, so an idiom can obscure what is asked but not what the model must know. Models near the chance floor also generate few sole-cause events (OLMo-2-1B yields $b = 8$, $c = 11$), leaving little power there.

MNLI: consistent idiom-specific degradation. MNLI is the largest evaluation and the one where the manipulation bears directly on the decision. Ten models show a significant penalty, from Falcon3-3B at −2.25 pp to Qwen2.5-7B at −1.17 pp (Table 2), and the sole-cause test agrees: $b > c$ for 15 of 17 models, both tests concurring at $p < 0.05$ for 9 of them — the same models bar Mistral-7B-v0.3, whose accuracy delta is significant while its sole-cause gap ($b = 225$, $c = 191$) is not. Agreement between two tests with different evidence bases across nine independent models is the strongest single piece of evidence we obtain, and it matches the design argument of §2: an entailment judgement turns on a literal reading of the premise, so a figurative surface adds a resolution step that can fail.

Robustness to generation-format failures. An unparseable generation is scored as incorrect, so a variant provoking more format violations is penalised for a reason unrelated to idiom comprehension. Rates are low on average (1.3% on SST-2, 0.7% on MMLU, 0.3% on MNLI) but uneven — StableLM-2-1.6B fails on up to 10.5% of MMLU items — and on SST-2 unbalanced across variants: the *paraphrase* arm fails more often than the idiomatic arm for 14 of 17 models, inflating Δ_{idi} . Dropping every triple containing a parse failure (the parse-clean columns of Table 1) leaves MMLU and MNLI untouched, all 10 significant MNLI models included, but halves the SST-2 mean from +1.31 to +0.67 pp and cuts its significant models from 5 to 3. Qwen2.5-7B is the clearest case: an apparent +3.04 pp idiom *advantage* becomes −0.21 pp, because its paraphrase arm failed to parse on 6.40% of rows against the idiomatic arm’s 2.07%. The direction does not change — even filtered, no SST-2 model shows a significant idiom penalty.

No capability or scale trend. Idiom sensitivity does not shrink as models grow. Spearman correlations between Δ_{idi} and original accuracy are −0.13 ($p = 0.62$) on SST-2, +0.18 ($p = 0.49$) on MMLU and −0.29 ($p = 0.26$) on MNLI; against parameter count, +0.09, +0.07 and −0.24 — all far from significance. Within the Qwen2.5 ladder on MNLI the effect is non-monotone (−0.40, −1.59, −0.85, −1.17 pp at 0.5B / 1.5B / 3B / 7B), and the two largest models in the panel sit inside the degraded group rather than above it. No correlation reaches significance in either direction; the nearest is a control rather than a treatment effect, SST-2 Δ_{par} against accuracy ($\rho = -0.43$, $p = 0.08$), whose sign is what a ceiling effect predicts. Rewriting is also not purely destructive — pooled across models, 526 SST-2, 284 MMLU and 1,957 MNLI rows were wrong on the original but right on both rewrites, which is precisely why the paraphrase control is needed.

4 Discussion

The sweep supports a narrower claim than the one usually assumed. Idiomatic phrasing is not a general accuracy tax on instruction-tuned decoder LLMs: on binary sentiment it did not hurt at all, and on knowledge QA it reached significance for only one model of seventeen. It appears instead to be a cost that materialises when the figurative surface is *load-bearing for the decision*, which is exactly the MNLI configuration. Where idioms can only obscure, the cost tracks how directly the rewritten span bears on the answer: MMLU rewrites the stem while the knowledge sits in untouched choices (-0.71 pp), MNLI rewrites the exact span the label is computed from (-1.16 pp). SST-2 does not sit on that scale at all but inverts it ($+1.31$ pp), its target property being one idioms tend to carry rather than obscure. That coupling, rather than the presence of idioms, is the most useful predictor we can offer.

The magnitudes are modest even at their strongest. A 1–2 pp mean drop on MNLI is real and statistically robust, but it is the same order as the cost of plain rewording rather than an order above it, so idiomaticity is better read as one distinguishable contributor to paraphrase sensitivity than as a separate failure mode. This tempers, without contradicting, the weaknesses reported by Tayyar Madabushi et al. [2], on idiomaticity detection, and Chakrabarty et al. [3], on figurative understanding, both measured directly; what is measured here is the downstream cost surviving after a model applies whatever partial understanding it has. Practically this yields a triage rule: work on idiom robustness pays off where figurative language can land on the span that determines the answer, and is largely wasted where the decision is carried elsewhere. The generic finding deserves equal weight, since on SST-2 plain paraphrasing cost more than idiomatizing did, echoing the brittleness to answer-preserving rewrites reported by Ribeiro et al. [5], though sought there by adversarial search rather than arising from neutral rephrasing. Least comfortable is the absence of a scale trend: across an 18-fold parameter range the effect neither shrinks nor grows reliably, so a larger checkpoint should not be expected to bring idiom robustness with it.

4.1 Threats to validity

Cross-dataset magnitudes carry an augmenter confound. MNLI was augmented with `gemini-3.5-flash-lite`, the other two with `gemini-3.6-flash`, and rewrite depth differs accordingly: median token-set overlap with the original is 0.33 on MNLI against 0.39 on SST-2 and 0.54 on MMLU, though dataset and augmenter vary together and cannot be separated post hoc. Within a dataset both arms share one augmenter, so the paraphrase control still absorbs rewrite depth and the contrast stays internally valid; what is not cleanly comparable is MNLI’s effect magnitude against the other two.

Validation is incomplete and not independent. Validation covers label preservation and idiom presence/absence, but nothing tests whether a rewrite preserved the original meaning, so a rewrite that drifted semantically while keeping the label plausible could survive — which matters most exactly where the effect was found, since an MNLI premise whose meaning shifts changes the entailment relation outright. A single client both writes and grades the rewrite, so validation is not independent of generation either.

The statistical treatment has two known weaknesses. The parse-failure filter is blunt — it drops 2.8% of SST-2 rows on average but over 12% for two models there, and 15.9% for StableLM-2-1.6B-Chat on MMLU, and dropping rows non-randomly can itself bias a comparison — so filtered and unfiltered readings bracket the truth. The pooled tests also treat models’ items as exchangeable when the same row recurs once per model; a mixed-effects model with model as a random effect was not run, so pooled figures — including MMLU’s, whose significance rests entirely on pooling — are indicative, not definitive.

Residual dataset artifacts. MNLI merges the matched and mismatched validation splits, mixing five genres absent from training with five present, and the length filter pruned genres unevenly (50% retention on fiction to 89% on `oup`); genre is recorded per row but not analysed. Forty MNLI paraphrase rows (0.85%) are byte-identical to their premise, weakening the control marginally. All three datasets

come from public splits that may appear in pretraining corpora, inflating absolute accuracy — though largely orthogonal here, since every reported quantity is a within-row difference.

Scope limits. Every model was evaluated zero-shot, greedily, under a single prompt template per dataset, so few-shot prompting, sampled decoding and prompt-format variation are untested and could interact with idiomatcity; the fine-tuned encoder track is deferred. Coverage has one hole: Llama-3.2-1B-Instruct is gated on Hugging Face and was never run.

Future work. The most valuable next step would sharpen the MNLI finding rather than broaden it: a direct check that rewrites preserve meaning, ruling out semantic drift as an alternative explanation. A mixed-effects re-analysis would then put the pooled claims on a proper footing, and extending to few-shot prompting would establish whether in-context examples recover the loss, as labelled examples do for the fine-tuned encoders of Tayyar Madabushi et al. [2].

5 Code

Code repository: https://github.com/LihuZur/Idiom_degradation

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